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Analysis and modeling of edge fluctuations and transport mechanism in the Maryland Centrifugal Experiment

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The Maryland Centrifugal Experiment [R. F. Ellis *et al.*, Phys. Plasmas **12**, 055704 (2005)] is a mirror machine designed to have a plasma axially confined by supersonic rotation and dominantly interchange stable by the radial shear in the azimuthal velocity. Nevertheless, residual fluctuations still persist. To investigate the presence of such fluctuations, an azimuthal array of 16 magnetic pickup coils at the edge region of the plasma has been employed. A comprehensive analysis of the magnetic fluctuations reveals that, under the imposed shear flow, only $m=0$ and $m=2$ modes are dominant; yet, the observed frequency spectrum is broadband. Using higher order spectral analysis, clear evidence of nonlinear mode coupling is detected. It is also observed that the amplification of magnetic fluctuations leads to enhanced transport consistent with the drop of the plasma density and voltage. As a result, the magnetic fluctuations start to decrease in amplitude as the central plasma pressure drops. In return, the anomalous radial particle and momentum transport are reduced; thus, the plasma confinement improves. As the plasma pressure starts to build up, the plasma voltage increases, destabilizing the $m=2$ interchange mode. The cycle of enhanced transport and intermittent fluctuations repeats itself. A two-dimensional magnetohydrodynamics code in slab geometry is employed to investigate the dynamics of the primary interchange instability and to assess the level of transport. For very low sheared rotation, a broad spatial spectrum of unstable modes is obtained. As the sheared rotation is increased, the high mode numbers become stabilized and low mode numbers dominate the spectrum. Both the experimental data obtained from the azimuthal array probes and the simulations in case of parabolic shear flow show clear evidence of nonlinear mode coupling, explaining the broadband frequency spectrum for low mode numbers. A detailed comparison of spatiotemporal dynamics of simulations with the experimental data is presented. © 2008 American Institute of Physics. [DOI: [10.1063/1.3028312](https://doi.org/10.1063/1.3028312)]

I. INTRODUCTION

A stable rotating mirror device relevant to thermonuclear fusion was first investigated in the 1970s.¹ The primary idea behind this configuration is the prevention of a loss cone with an axial component of the centrifugal force, which confines the plasma away from the mirror throats. These plasmas are assumed to be well confined in the direction transverse to the magnetic field. However, rotating plasmas can become interchange unstable due to a combination of the radial outward centrifugal force and the curvature in the magnetic field within a mirror geometry. Recently, in toroidal magnetic confinement devices, it has been established that radial shear in poloidal or toroidal flows can effectively suppress unstable modes that give rise to anomalous radial transport.^{2–4} Thus, it becomes clear that if the azimuthal rotation in rotating plasmas with mirror geometry has sufficient shear, it may suppress interchange modes. A combination of improved axial confinement by the rotation as well as improved radial confinement due to the shear flow has led to a renewed interest in rotating plasmas in mirror geometries.^{5,6} The Maryland Centrifugal Experiment (MCX) has been developed as an alternative approach to conventional toroidal field configuration devices for the purpose of thermonuclear fusion. Indeed, various rotating plasma experiments have recently been con-

structed to investigate confinement in the presence of shear flow.^{7–12}

To achieve an improved confinement in mirror geometry, plasma must rotate at sonic Mach numbers ~ 4 – 5 . In the Maryland Centrifugal Experiment, sonic Mach numbers between 2 and 3 have been achieved.⁶ Thus, it is anticipated that the resulting shear flow may not be adequate to completely suppress all the unstable interchange modes. The characteristics of the residual interchange fluctuations has been the focus of recent experimental studies on MCX.¹³ In this paper, detailed observations of the spatial and temporal characteristics of these fluctuations are presented. These results are also compared with two-dimensional magnetohydrodynamics (2-D MHD) simulations of a rotating plasma with plasma parameters similar to those in the experiment.

II. EXPERIMENTAL SETUP

MCX has a magnetic configuration of a long solenoid with axisymmetric mirror ends. The distance between two mirror throats is 2.5 m and the maximum field at the mirror throat can be as high as 2 T. The midplane field is controlled independently to allow us to vary the mirror ratio. MCX uses an axial magnetic field and a radial voltage to drive supersonic azimuthal flows. The inner electrode is a stainless steel core (4.8 cm diameter) along the axis of the machine. The

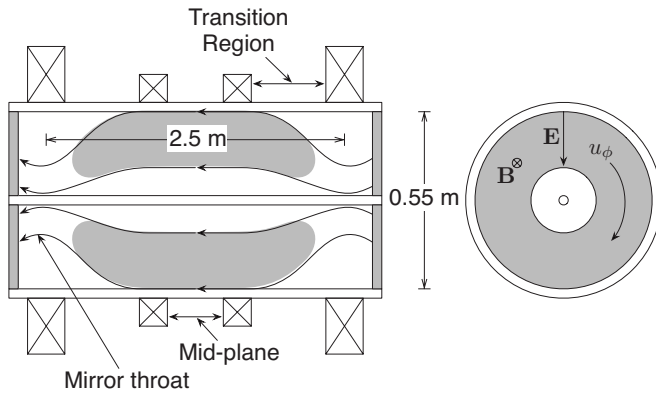


FIG. 1. Schematic of MCX.

core is insulated at mirror throats using high purity alumina sleeves. The machine is initially filled with hydrogen gas; biasing the core with negative voltage with respect to the vacuum vessel generates the plasma discharge and $\mathbf{E} \times \mathbf{B}$ azimuthal rotation in the plasma region (Fig. 1). The azimuthal rotation gives rise to an axial force which confines the plasma in the midplane; moreover, it has a strong radial shear that can stabilize the interchange modes (Fig. 2).

For the experimental results presented in this paper, MCX is initially filled with hydrogen gas up to 8 mTorr, and the core is biased to -10 kV using an ignitron switched capacitor bank. The MCX plasma reaches a steady state at around 1 ms after the initial discharge switch. The plasma is usually terminated after about 5 ms by using a crowbar ignitron short circuiting the plasma to the ground.

During the plasma discharge, plasma voltage and current are monitored using a voltage probe and a Rogowski coil on the core. Based on the zero-dimensional circuit model, equivalent stored charge in the plasma can be directly measured by integrating the current reversal at the time of the crowbar ignitron fire. An infrared laser interferometer with wavelength $\lambda = 3.39 \mu\text{m}$ is employed to measure the line averaged electron density at the midplane. It is well documented that the measured MCX plasma density is in good agreement with the volume averaged plasma density calculated from the stored charge.⁶

In order to calculate the average plasma velocity for $\mathbf{E} \times \mathbf{B}$ azimuthal rotation, as seen in Fig. 3(b), we use the time traces of the load voltage V_p , and the cross-field extent of the plasma, $a = 0.2$ m, interpreted from the vacuum configuration of the magnetic field, $B = 2$ kG, at a known mirror ratio, $R_m = 7$. Average $\mathbf{E} \times \mathbf{B}$ drift velocity calculated using the for-

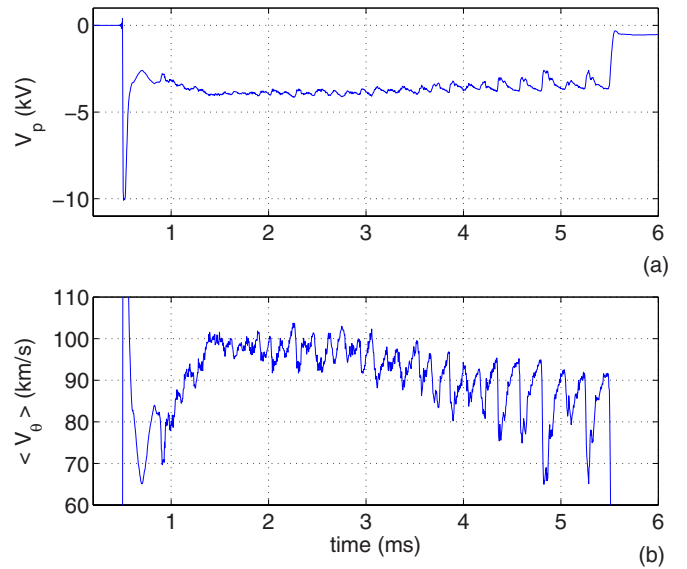


FIG. 3. (Color online) (a) Plasma load voltage from the core. (b) Calculated average rotation velocity.

mula V_p/aB , has been proven to be in good agreement with the measured Doppler resolved, parabolic radial velocity profile.¹⁴ Decay in the load voltage is correlated with the plasma rotation velocity slowing down, thereby losing the confinement.

In addition to these basic diagnostics, 16 magnetic pickup coils placed uniformly in an azimuthal array at the edge region of the plasma are used to measure the magnetic fluctuations in the axial direction (Fig. 4). The location of the azimuthal array is halfway between the midplane and the mirror throat at a radius of 0.24 m. Because of their uniform separation in the azimuthal direction, the resolution of mode numbers up to $m=8$ is obtained. Each pickup coil is about 1 cm in diameter and isolated via a nonconducting heat shield, then molded into a stainless steel encasing which acts as an electrostatic shield. The calibration of the probes has been done using a Helmholtz coil and has been done in situ using the MCX magnetic field. The calibration curve is flat up to about 12 kHz (consistent with L/R time for the pickup coils) with subsequent rollover as a function of frequency. The cutoff frequency is observed to be near 500 kHz. In this paper, we present observed spectral results and time history of the magnetic fluctuations and we will address the ob-

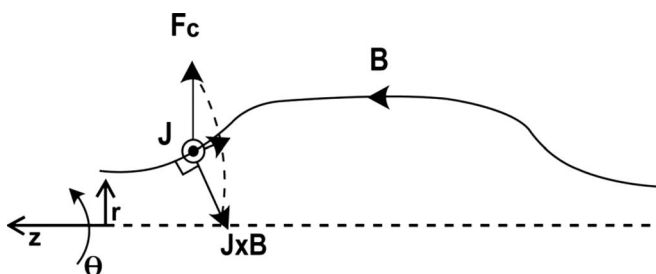


FIG. 2. Axial confinement force in MCX.

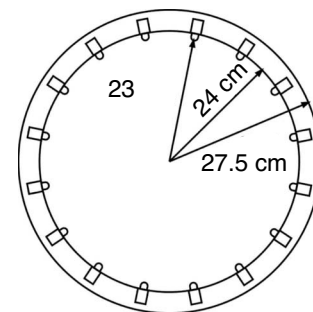


FIG. 4. Azimuthal array of pickup coils.

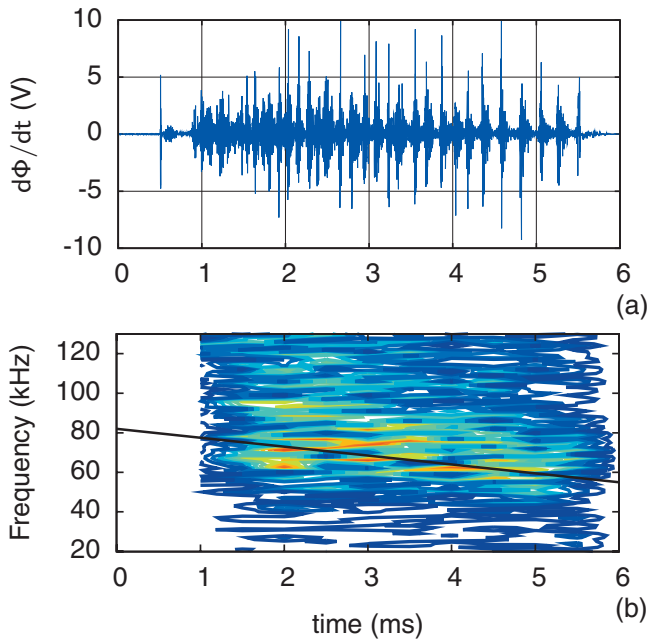


FIG. 5. (Color) (a) Voltage reading from one of the pickup coils, $d\Phi/dt$, where Φ is the magnetic flux. (b) The spectrogram results for one of the pickup coils. Solid line is used to indicate the shift in the peak frequency.

served quadratic mode coupling and transport mechanisms by means of 2D magnetohydrodynamic simulations.

III. RESULTS

A. Analysis

As a part of standard shot procedure, the data acquisition system is triggered 0.5 ms prior to the discharge. The sampling rate is set to 1 MS/s, providing 1 μ s time resolution. The plasma formation phase lasts about 1 ms after the start ignitor (see the voltage and velocity traces in Fig. 3). In Fig. 5(a) is shown the signal obtained from one of the pickup coils. Due to possible transient events, it is impossible to employ an ensemble averaging over the whole duration of the plasma. In order to identify a time interval of similar events, a straightforward approach would be to perform the short-time Fourier transform (STFT) on a block by block basis, using a window function for consecutive time intervals with 50% overlap.¹⁵ Each resulting Fourier transform describes the signal's frequency distribution in the vicinity of the center of the window function; yet, by moving the window function, and repeating the same process, this method offers a rough idea of the time evolution of the signal's frequency content. Often, this application of the short-time Fourier transform is called the "spectrogram" in signal processing. For a standard MCX shot, the STFT is employed over 1 ms time intervals, i.e., 1000 samples, for the azimuthal array probe data [Fig. 5(b)]. The spectrogram plot reveals that after the plasma formation phase, starting at around $t = 2$ ms, the dominant plasma fluctuations have many components within a broad peak with frequency spread comparable to its central frequency $f = 70$ kHz. It also observed that this broad structure shifts toward lower frequencies at later time, confirming the slower rotation velocity observed in Fig. 3(b).

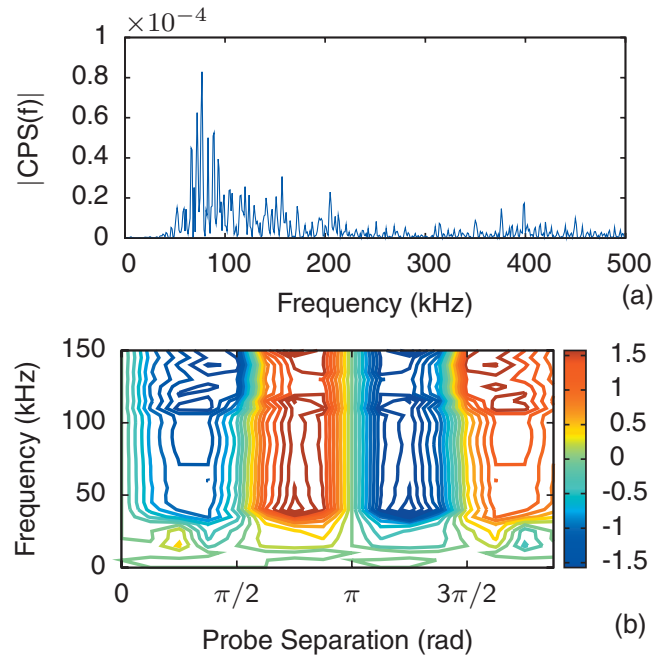


FIG. 6. (Color) (a) The amplitude of the cross-power spectrum calculated from the time traces of two pickup coils separated by 30 cm. (b) The phase spectrum obtained using all possible separations for 16 pickup coils.

Using the spatial Fourier transform of 16 probes on the array, mode numbers for the magnetic fluctuations up to $m = 8$ can be identified. In the two-dimensional Fourier transform, results show that the data are turbulent in the frequency domain yet low dimensional in space, defined with only two different mode numbers. There is an $m=0$ mode with frequency around 6 kHz, and an $m=2$ mode with a broad spectrum between 50 and 120 kHz.¹⁶ These results are reproducible for each plasma shot, even though the peak of the frequency distribution and the low frequency component may vary from shot to shot due to a slight change in the $\mathbf{E} \times \mathbf{B}$ rotation velocity.

Another approach to identify mode structure is to calculate the cross-power spectrum (CPS) using the data from two pickup coils that are separated by a known distance.¹⁷ Let $X(f)$ and $Y(f)$ be the Fourier transform of the two fluctuation signals $x(t)$ and $y(t)$ obtained from two pickup coils separated by a known distance. The definition of the cross-power spectrum is

$$\text{CPS}(f) = \langle X(f)Y^*(f) \rangle = |X(f)||Y(f)|e^{i\theta(f)}. \quad (1)$$

The amplitude of the result is used to analyze the frequency distribution of the fluctuations, whereas $\theta(f)$ denotes the phase spectrum. Using the 16 pickup coils, we can plot the phase spectrum as a function of separation. This algorithm is employed for the time interval $2 \text{ ms} \leq t \leq 3 \text{ ms}$, assuming a steady state. In Fig. 6(a), the amplitude of the cross-power spectrum for a pair of pickup coils separated by 0.3 m is plotted. In Fig. 6(b) are shown the results for the phase spectrum as a function of probe separation. The phase spectrum results also show spatial periodicity for frequencies from 50 to 120 kHz, and a constant phase for low frequencies from 0 to 30 kHz, confirming the two-dimensional Fourier

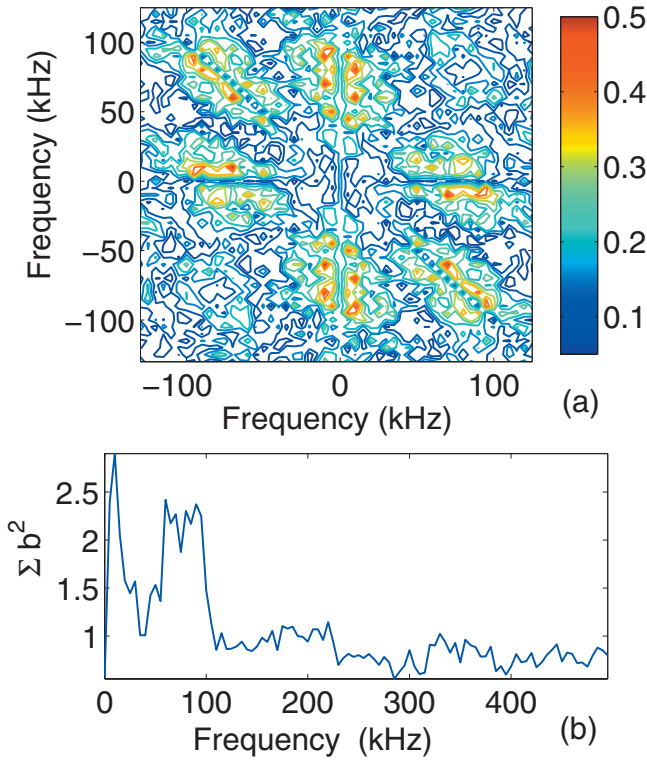


FIG. 7. (Color) (a) The cross-bicoherence results calculated using a pair of pickup coil signals separated by 0.2 m. (b) The summed bicoherence results.

transform results. Furthermore, the amplitude of the cross-power spectrum shows that the various sidebands within the broadband structure are separated by 6 kHz, indicating a possible nonlinear mode coupling between the $m=2$ mode and the low frequency $m=0$ mode.

It is well established that the bispectrum and its phase spectrum, i.e., the bicoherence, can be used as a method for investigating possible evidence of quadratic nonlinear mode coupling.^{18–20} $X(f)$ and $Y(f)$ being the fast Fourier transform of a pair of probe signals measured simultaneously, i.e., $x(t)$ and $y(t)$, the bicoherence is defined as

$$b_{ij} = \frac{\langle X_i Y_j X_{i+j}^* \rangle}{\sqrt{|X_i|^2 |Y_j|^2 |X_{i+j}|^2}}, \quad (2)$$

where i and j are the indices for each frequency component. In Fig. 7(a) is seen the result of the cross-bicoherence calculations assuming a steady state for $2 \text{ ms} \leq t \leq 3 \text{ ms}$ time interval and probe separation of 0.2 m. It should also be noted that no changes in the level of coherence is observed for various probe pairs.

Often, for simplicity, the summed bicoherence $\sum_j b_{ij}^2$ is employed instead of the bicoherence. For the experimental observations, the summed bicoherence plot shows that the broadening of the high frequency component is indeed a result of strong nonlinear coupling to a low frequency mode at 6 kHz ($m=0$) [Fig. 7(b)].

In Fig. 8 is shown the density obtained using the infrared laser interferometer, the signal from one of the pickup coils, and the plasma voltage as a function of time over a 800 μs period. As the magnitude of the plasma voltage drops (m

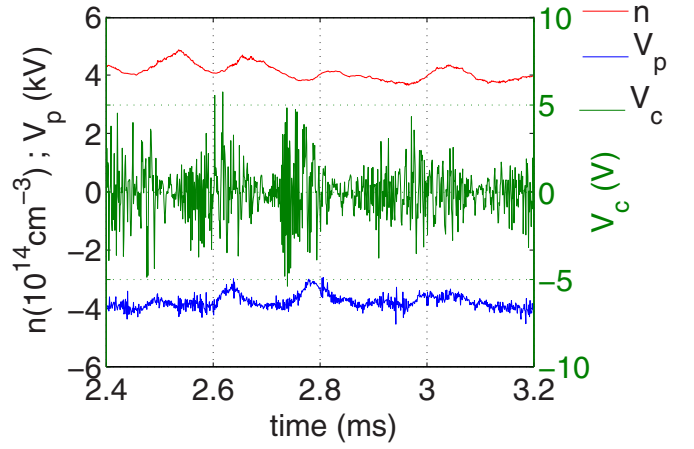


FIG. 8. (Color online) The measured density (n), the load voltage (V_p), and the raw signal from one of the pickup coils (V_c) as a function of time over a 800 μs period. The density has a unit of 10^{14} cm^{-3} , the load voltage is in kilovolts, and the pickup coil signal is in volts.

$=0$), a sharp decrease in the measured density is observed. On the other hand, data from the pickup coils measuring the rate of change of magnetic flux ($d\Phi/dt$) show that the buildup in the magnetic fluctuations precedes the drop in the load voltage. As the load voltage reaches its minimum, the fluctuations are strongly reduced. As the plasma density and voltage increase, the fluctuations build up again followed by a drop in the voltage.

It is clear that the nonlinear mode coupling plays a crucial role between the states of enhanced transport and improved confinement. The applied voltage leads to a shear flow which suppresses the fluctuations from the interchange modes. In return the anomalous radial particle and momentum transport are reduced; thus, confinement is improved. As a result, the central plasma pressure builds up and also the plasma flow (voltage) increases. This increase in the radial pressure gradient destabilizes the $m=2$ interchange mode. The amplification of this mode then leads to enhanced transport. As the voltage and the plasma density reach their minimum due to enhanced transport, the pressure drops, thus the drive for the instability is reduced. Consequently, the fluctuations decrease in amplitude, and the cycle repeats itself.

B. 2D MHD model

To model the nonlinear development of interchange modes, we have performed numerical simulations based on MHD equations solved in a two-dimensional slab geometry for an isothermal plasma, in the presence of radial parabolic driven azimuthal shear flow. The centrifugal force, which arises from both the rotation of the plasma as well as the curvature of the field lines, is modeled using an effective gravity:

$$\frac{\partial n}{\partial t} + \vec{\nabla} \cdot (n\vec{u}) = S, \quad (3)$$

$$\frac{\partial(nM\vec{u})}{\partial t} + \vec{\nabla} \cdot (nM\vec{u}\vec{u}) = -\vec{\nabla} \cdot \left(2nT + \frac{B^2}{8\pi} \right) + \frac{\vec{B}}{4\pi} \cdot \vec{\nabla} \vec{B} + \vec{\nabla} \cdot (nM\mu \nabla \vec{u}) + \vec{F}, \quad (4)$$

$$\frac{\partial \vec{B}}{\partial t} = \vec{\nabla} \times (\vec{u} \times \vec{B}) + \eta \frac{c^2}{4\pi} \nabla^2 \vec{B}. \quad (5)$$

The term S on the right hand side of Eq. (3) is the source/sink term maintaining the equilibrium density profile. The constant source term $\vec{F}$ in Eq. (4) drives a parabolic azimuthal flow for no-slip boundary conditions. These equations are solved in two dimensions in a slab geometry with x being the radial direction and z , the periodic direction, representing the azimuthal direction in the MCX device. For this case, only the B_y field is evolved. The chosen normalization is $x \rightarrow x/a$, $z \rightarrow 4\pi z/a$, $t \rightarrow t\sqrt{g/a}$, $n \rightarrow n/n_0$, $B \rightarrow B/B_0$, $T \rightarrow T/T_0$, and $u \rightarrow u/c_s$, where c_s is the sound speed. In the simulations $T=0.1$ and $F=-2.0$. For conversion to real units, $t_0 = \sqrt{a/g} = 0.4 \mu s$, $n_0 = 4.5 \times 10^{14} \text{ cm}^{-3}$, $B = 0.2 \text{ T}$, $a = 0.15 \text{ m}$. The number of grid points chosen for the simulations presented in this paper are $N_x=30$ and $N_z=90$. The boundaries are defined as 0 and 2π for the periodic direction (z) and -15 to 0 for the radial direction (x), -15 indicating the edge region, and 0 being the core.

In Fig. 9(a) is shown the equilibrium two-dimensional density profile and in Fig. 9(b) is shown the equilibrium velocity profile for $F=-2$. The maximum Mach number in the steady state without fluctuations is 4. The simulations are performed in a half-space (in x) problem. This is accomplished by setting Neumann boundary conditions for the flow and the density in the x direction at the right boundary. Since there is no variation in z , the density is convected by the flow in that direction, where all dependent variables satisfy periodic boundary conditions.

The first case studied will be one in which there is no azimuthal flow. Random noise with an amplitude of 0.02 is added to each grid point using a random number generator uniformly distributed between 0.02 and -0.02 . The fluctuations driven by the interchange modes in case of no shear flow grow very rapidly within a very short time. Figures 10(a) and 10(b) are plotted for $t=20$. As seen in Fig. 10(a), the density profile shows that the fluctuations are radially elongated indicating high azimuthal mode numbers as expected. In Fig. 10(b) is shown the self-consistent azimuthal flow associated with the interchange modes which also grow to large levels. The low density from the edge penetrates all the way into the center; the high density starts spreading to the edge leading to rapid transport. Even though there is a source maintaining the density profile, the “anomalous” radial particle transport driven by the interchange modes is so large that it effectively flattens the density profile at late time. These all occur on a time scale of about 100 ideal growth times. Finally, at much later time the residual instability is at lower mode numbers and the amplitude of fluctuations is small since the primary drive for the instability, namely, the density gradient, has been eliminated.

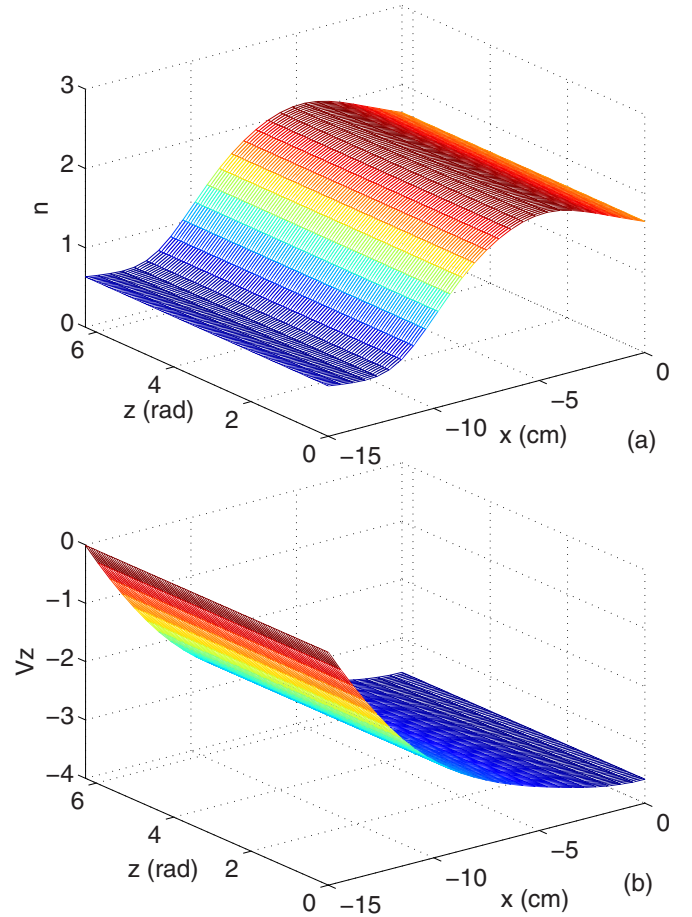


FIG. 9. (Color online) Equilibrium profiles obtained using simulations. Density profile (n) is normalized to $4.5 \times 10^{14} \text{ cm}^{-3}$ and the azimuthal velocity (V_z) is normalized to sound speed c_s .

As stated earlier, to generate the azimuthal flow of Mach number of 2, which is observed in the experiments, the forcing term F is set to a constant value of -2 . The negative sign is chosen to give an azimuthal flow consistent with the experimentally observed direction for the flow. Shown in Figs. 11(a) and 11(b) are the 2D density and azimuthal flow structures. What is obvious is that the density profile has been re-established. However, the fluctuation levels for the flow and the density are quite high especially in the core where, due to the parabolic nature of the flow, the shearing effect is weak, hence, the stabilization of the interchange modes is not effective. In Fig. 9(b), the Mach number is about 2.5 times larger than in Fig. 11(b) for the same value of the forcing amplitude. This is because the fluctuations give rise to anomalous momentum transport which reduces the average equilibrium flow. One key difference compared with the simulations without flow (Fig. 10) is that the fluctuations are now at much lower mode numbers.

To make a more quantitative comparison with the experimental data, in the edge region of the simulation box, at 16 locations in z , $\pi a/8$ apart, the time history of the magnetic field component B_y , which is the same direction measured by the pickup coils, is recorded and then analyzed. Using the experimental data from the 16 pickup coils, the magnetic field is computed by a simple integration and nor-

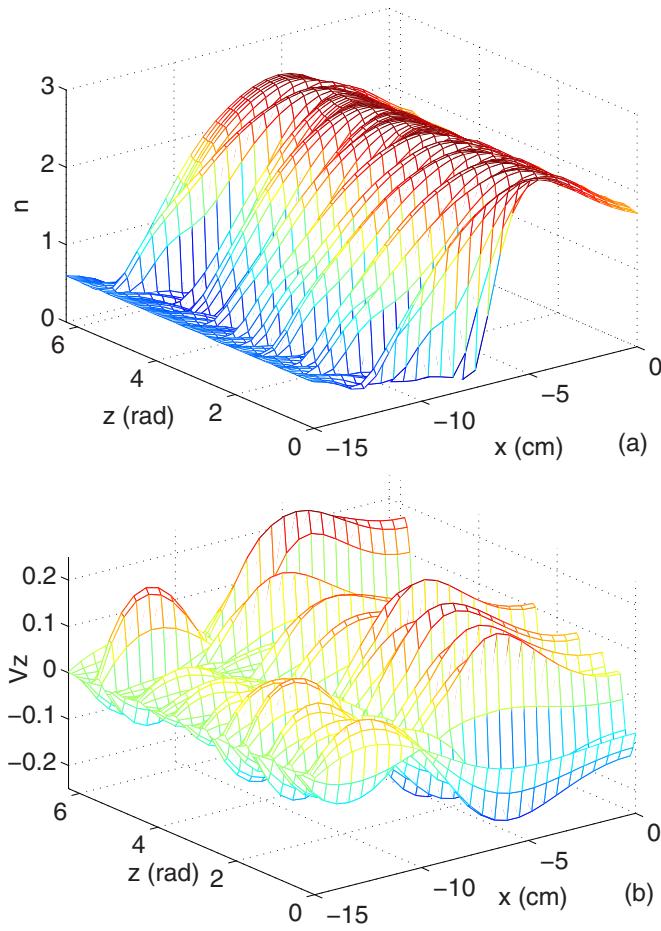


FIG. 10. (Color online) Simulated density and velocity profiles without flow, $F=0$. Again, the density is normalized to $4.5 \times 10^{14} \text{ cm}^{-3}$ and V_z is in units of sonic Mach number (M_s).

malization discussed earlier.¹³ The results for the magnetic field are shown in Fig. 12(a) as a 2D contour plot of $I(m, t)$, where I is the intensity in each of the harmonics and t is the time. As expected, the azimuthal spectrum is dominated by an $m=2$ component which has a bursty, intermittent character. A similar plot obtained from the simulation data is shown in Fig. 12(b). A comparison of these plots shows that the bursting feature of the $m=2$ fluctuations is clearly a common feature. However, unlike the experimental observations, the simulations also have an $m=1$ component. Both plots show that the levels of fluctuations in the higher modes are either absent or very small.

For further quantitative comparison, in Fig. 13(a) is plotted the cross-bicoherence computed using the numerical data from the simulated pickup coils' signals. There is a very good agreement with the observed cross-bicoherence shown in Fig. 7(a) and that from the simulations. However, due to the presence of the $m=1$ mode in the simulations, there are additional features in the bicoherence plot indicating quadratic mode coupling between $m=0$ and $m=1$ modes. The presence of the $m=1$ fluctuations makes the cross-bicoherence plot accentuating the broadening of frequency components of $m=1$ mode, unlike that in the observed cross-bicoherence where the quadratic coupling is only between the low frequency from the $m=0$ and the broadband high

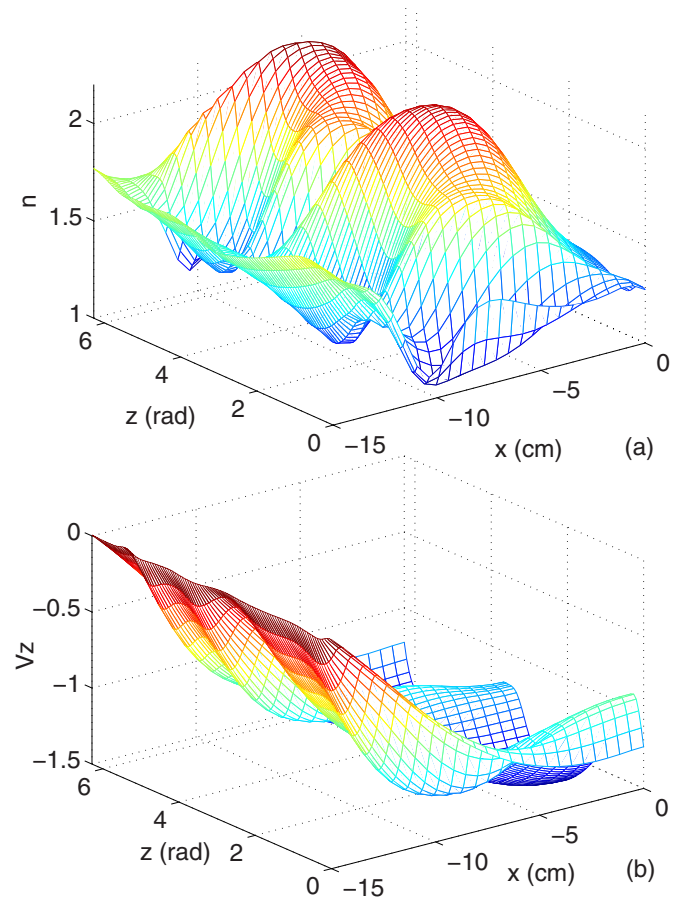


FIG. 11. (Color online) Simulated density and velocity profiles with flow, $F=-2$. The azimuthal velocity V_z is normalized to sound speed (in units of M_s) and the density profile n is normalized to $4.5 \times 10^{14} \text{ cm}^{-3}$.

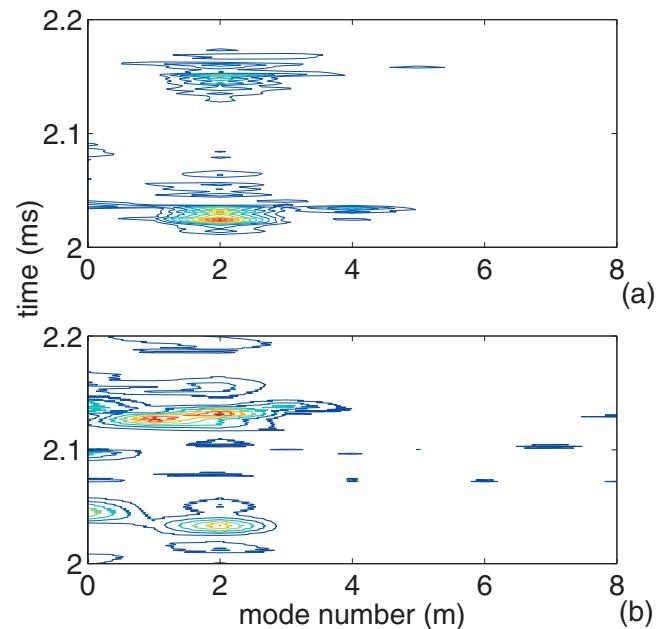


FIG. 12. (Color) (a) Spatial Fourier transform analysis of the observed magnetic fluctuations. (b) Spatial Fourier transform analysis of the simulated magnetic fluctuations with flow, $F=-2$.

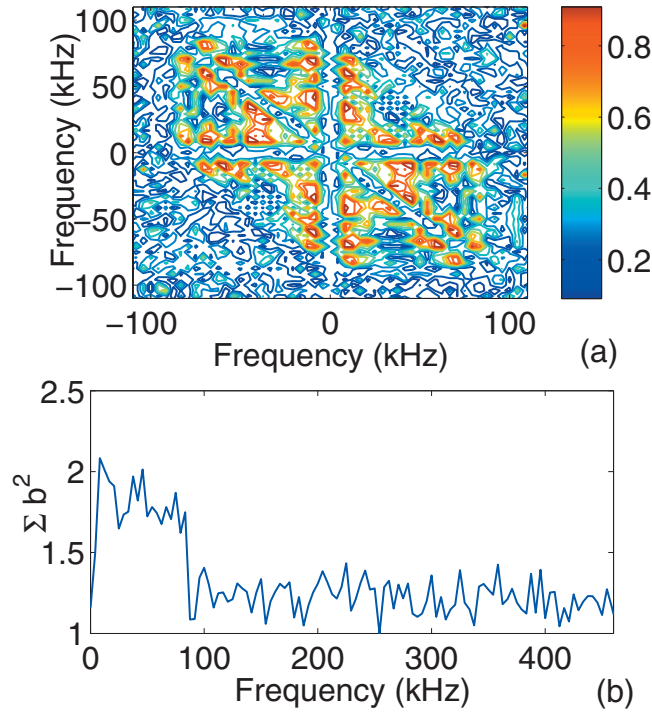


FIG. 13. (Color) (a) The cross-bicoherence result calculated using numerical simulations of the pickup coil signals separated by $0.2 m$. (b) The summed bicoherence result for the simulated pickup coil signals.

frequency $m=2$. Similarly in the summed bicoherence results for the simulations with flow, displayed in Fig. 13(b), due to the presence of the $m=1$ fluctuations, the spectrum is filled in between the low frequency $m=0$ and the high frequency $m=2$, unlike the distinct low frequency peak and the broader yet separate high frequency peak clearly seen in the data [Fig. 7(b)]. Nevertheless, even with this difference the agreement is very impressive.

Finally, in Fig. 14 is plotted a trace of the density at a core position, the plasma load voltage, computed by integrating the azimuthal velocity in the x direction for each point in z and then averaging in z to get the $m=0$ component, and

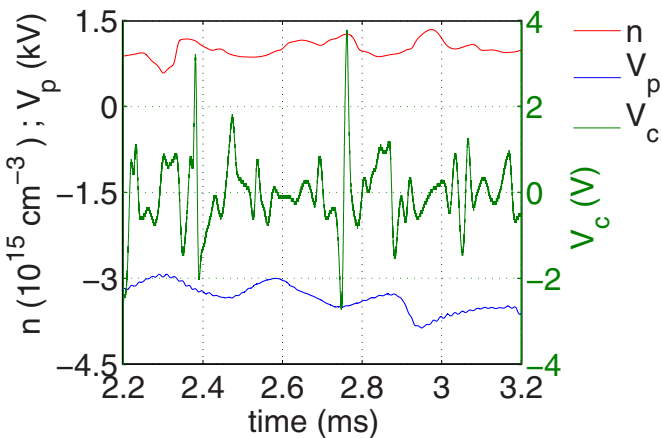


FIG. 14. (Color online) Simulated density, load voltage, and pickup coil signals as a function of time over a 1 ms period. Density has a unit of 10^{15} cm^{-3} . The blue (bottom) solid line is the load voltage trace in kilovolts. The green (middle) line represents the simulated pickup coil signal (V_c) in volts.

voltage computed from rate of change of magnetic flux to mimic the pickup coil voltage from one of the azimuthal locations, as a function of time using the simulations with flow, $F=-2$. The drop in the core density is clearly seen to be correlated with the drop in the voltage. This is again consistent with the behavior seen in the experimental data displayed in Fig. 8. The drop in the load voltage and the density is a signature of the particle and momentum loss. The comparison of the pickup coil voltage between the data and the simulations, however, is not very good. The observations in Fig. 8 clearly show the buildup in the high frequency fluctuations on the pickup coil which occurs when the density and load voltage are increasing. When the fluctuations reach a substantial level, they trigger a loss in the number of particles and the azimuthal momentum which accounts for the drop in the density and the load voltage. This drop flattens the drive for the interchange instability leading to a reduction in the magnetic fluctuations, a subsequent improvement in the confinement and a repetition of this cycle. This simple causal relationship is very clearly seen in the observed data but not so in the simulations. This again is perhaps tied in with the occurrence of the $m=1$ mode, which does not allow for a clear separation in time scale for the $m=0$ activity which epitomizes the transport and the $m=2$, for which there are much higher frequency fluctuations. Namely, there is a clear distinction in the transport time scale of about $100 \mu\text{s}$ compared to the fluctuation time scale of $5-10 \mu\text{s}$ seen in the data. However, in the simulations with the occurrence of the $m=1$ structure, clear separation between these two time scales is obliterated by this activity occurring at intermediate time scales.

IV. CONCLUSION

Here we have presented the analysis of observed magnetic fluctuations from the rotating mirror plasma MCX with sheared flow. The magnetic fluctuations measured by the sixteen pickup coils show that the fluctuations are broadband in frequency even though they are low dimensional in space with only $m=0$ and $m=2$ modes. Both data analysis and simulations show that there is a strong nonlinear interaction between these modes. 2D MHD simulations indicate that in the presence of an imposed parabolic shear flow, only low mode numbers are unstable which intermittently get suppressed, thereby improving particle and momentum confinement ($m=0$). During this well-confined phase, the central plasma pressure builds up, thereby triggering only low mode number interchange modes. This is followed by periods of enhanced particle and momentum transport reducing the pressure gradient. The cycle of intermittent fluctuations and enhanced transport followed by periods of quiescence repeats itself. The strong interplay between the $m=0$ and $m=2$ modes leads to the broadband frequency spectrum observed for the $m=2$ mode. The simulation data, however, show that there is an additional $m=1$ mode present. Nevertheless, the 2D simulations show that the dynamics are dominated by low mode numbers. One, however, sees from the simulations that the fluctuation levels are quite large, especially in the core where the shear stabilization is not that

effective and in order to get better reduction of the fluctuations one needs to go to higher Mach number flows.

Reduction of fluctuations and restoration of density profiles to the laminar profile has been shown in simulations of velocity shear stabilized Z-pinch plasmas²¹ (Mach numbers of up to 4.8 were needed). In addition, we note that the present simulation is done in slab geometry and isothermal conditions with the main intent to demonstrate the turbulent condensation to long wavelengths. In cylindrical geometry, there is an additional physics aspect that could mitigate fluctuations further;²² namely, in mirror geometry, the field line curvature is destabilizing, being proportional to the pressure gradient ($-dp/dr$), whereas the centrifugal force itself is also destabilizing, being proportional to the density gradient ($-dn/dr$); the latter, however, is longer, by M_s^2 . This raises the possibility that the cylindrical system could confine pressure and be interchange stable by a weak outwardly increasing density.

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